Year 11 Foundation

REVISION PACK

Graphs 5

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 5 · Simultaneous equations (Lesson 3.1)

Q	Working	Answer
adding the equations		
1.1	$x + y = 12$ $x - y = 4$ $\text{add}: \ 2x = 16$ $x = 8$ $1 \times 8 + y = 12$ $y = 4$	$x = 8, y = 4$
1.2	$x + y = 11$ $x - y = 7$ $\text{add}: \ 2x = 18$ $x = 9$ $1 \times 9 + y = 11$ $y = 2$	$x = 9, y = 2$
1.3	$x + y = 12$ $x - y = 2$ $\text{add}: \ 2x = 14$ $x = 7$ $1 \times 7 + y = 12$ $y = 5$	$x = 7, y = 5$
1.4	$x + y = 13$ $x - y = 7$ $\text{add}: \ 2x = 20$ $x = 10$ $1 \times 10 + y = 13$ $y = 3$	$x = 10, y = 3$

1.5	$x + y = 7$ $x - y = 5$ $\text{add}: \ 2x = 12$ $x = 6$ $1 \times 6 + y = 7$ $y = 1$	$x = 6, y = 1$
1.6	$x + y = 15$ $x - y = 7$ $\text{add}: \ 2x = 22$ $x = 11$ $1 \times 11 + y = 15$ $y = 4$	$x = 11, y = 4$
subtracting the equations		
1.7	$2x + y = 13$ $x + y = 8$ $\text{subtract}: \ x = 5$ $2 \times 5 + y = 13$ $y = 3$	$x = 5, y = 3$
1.8	$4x + y = 14$ $x + y = 5$ $\text{subtract}: \ 3x = 9$ $x = 3$ $4 \times 3 + y = 14$ $y = 2$	$x = 3, y = 2$
1.9	$3x + y = 19$ $x + y = 7$ $\text{subtract}: \ 2x = 12$ $x = 6$ $3 \times 6 + y = 19$ $y = 1$	$x = 6, y = 1$

1.10	$5x + y = 17$ $x + y = 9$ subtract : $4x = 8$ $x = 2$ $5 \times 2 + y = 17$ $y = 7$	$x = 2, y = 7$
1.11	$2x + y = 20$ $x + y = 12$ subtract : $x = 8$ $2 \times 8 + y = 20$ $y = 4$	$x = 8, y = 4$
1.12	$3x + y = 21$ $x + y = 13$ subtract : $2x = 8$ $x = 4$ $3 \times 4 + y = 21$ $y = 9$	$x = 4, y = 9$
solving using graphs		
1.13	where the lines cross $(3, 5)$	$x = 3, y = 5$
1.14	where the lines cross $(2, 3)$	$x = 2, y = 3$
1.15	where the lines cross $(4, 3)$	$x = 4, y = 3$
1.16	where the lines cross $(2, 4)$	$x = 2, y = 4$
adding with a number in front of x		

1.17	$2x + y = 12$ $x - y = 3$ $\text{add}: \ 3x = 15$ $x = 5$ $2 \times 5 + y = 12$ $y = 2$	$x = 5, y = 2$
1.18	$4x + y = 13$ $x - y = 2$ $\text{add}: \ 5x = 15$ $x = 3$ $4 \times 3 + y = 13$ $y = 1$	$x = 3, y = 1$
1.19	$3x + y = 11$ $x - y = -3$ $\text{add}: \ 4x = 8$ $x = 2$ $3 \times 2 + y = 11$ $y = 5$	$x = 2, y = 5$
1.20	$5x + y = 11$ $x - y = 1$ $\text{add}: \ 6x = 12$ $x = 2$ $5 \times 2 + y = 11$ $y = 1$	$x = 2, y = 1$
1.21	$2x + y = 15$ $x - y = 3$ $\text{add}: \ 3x = 18$ $x = 6$ $2 \times 6 + y = 15$ $y = 3$	$x = 6, y = 3$

1.22	$4x + y = 10$ $x - y = -5$ $\text{add}: \ 5x = 5$ $x = 1$ $4 \times 1 + y = 10$ $y = 6$	$x = 1, y = 6$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Q	Working	Answer
equations with no solution		
2.1	Both gradients are 3, so the lines are parallel	**No solution**
2.2	Gradients 4 and -1 are different, so the lines meet $4x + 2 = -x + 7$, $x = 1$, $y = 6$	**Yes ($x = 1$, $y = 6$)**
2.3	Both gradients are -2, so the lines are parallel	**No solution**
2.4	Gradients 1 and -1 are different $x + 6 = 6 - x$, $x = 0$, $y = 6$	**Yes ($x = 0$, $y = 6$)**
2.5	$\frac{1}{2} = 0.5$, so both gradients are the same	**No solution**
2.6	$x + y = 5$ is $y = -x + 5$ Both gradients are -1, so the lines are parallel	**No solution**